

INVERSE TRIGONOMETRY LEVEL 1

NCERT / BOARD PATTERN – 150 QUESTIONS

Coverage: Basic definitions, principal values, identities, compositions, simple equations

SECTION 1: BASIC DEFINITIONS & PRINCIPAL VALUES (Q1–Q20)

1. Define an inverse trigonometric function.
2. Explain what is meant by the principal value of an inverse trigonometric function.
3. What is meant by $y = \sin^{-1} x$? State the corresponding relation between x and y .
4. What is meant by $y = \cos^{-1} x$? State the corresponding relation between x and y .
5. What is meant by $y = \tan^{-1} x$? State the corresponding relation between x and y .
6. State the domain and range of $\sin^{-1} x$.
7. State the domain and range of $\cos^{-1} x$.
8. State the domain and range of $\tan^{-1} x$.
9. On which interval is $\sin x$ restricted in order to define $\sin^{-1} x$ as a function?
10. On which interval is $\cos x$ restricted in order to define $\cos^{-1} x$ as a function?
11. On which interval is $\tan x$ restricted in order to define $\tan^{-1} x$ as a function?
12. Write the principal value ranges for $\cot^{-1} x$, $\sec^{-1} x$, $\csc^{-1} x$.
13. Why can $\sin^{-1} x$ be considered a one-one and onto function from $[-1,1]$ to $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$?
14. Why is $\cos^{-1} x$ defined from $[-1,1]$ to $[0,\pi]$?
15. Explain why the trigonometric functions must be restricted to suitable intervals to admit inverses.
16. What is the principal value of $\sin^{-1} x$ when $x \in [-1,1]$?
17. What is the principal value of $\cos^{-1} x$ when $x \in [-1,1]$?
18. What is the principal value of $\tan^{-1} x$ when $x \in \mathbb{R}$?
19. Give one example to show that $\sin^{-1}(\sin x)$ is not always equal to x .
20. Give one example to show that $\cos^{-1}(\cos x)$ is not always equal to x .

SECTION 2: SIMPLE EVALUATION PROBLEMS (Q21–Q50)

21. Evaluate $\sin^{-1} 0$.
22. Evaluate $\sin^{-1} \left(\frac{1}{2}\right)$.
23. Evaluate $\sin^{-1} \left(\frac{\sqrt{3}}{2}\right)$.
24. Evaluate $\sin^{-1} \left(-\frac{1}{2}\right)$.
25. Evaluate $\cos^{-1} 1$.
26. Evaluate $\cos^{-1} \left(\frac{1}{2}\right)$.
27. Evaluate $\cos^{-1} \left(-\frac{1}{2}\right)$.
28. Evaluate $\cos^{-1} 0$.
29. Evaluate $\tan^{-1} 0$.
30. Evaluate $\tan^{-1} 1$.
31. Evaluate $\tan^{-1} \sqrt{3}$.
32. Evaluate $\tan^{-1} (-1)$.
33. Evaluate $\cot^{-1} 1$ in its principal value.
34. Evaluate $\sec^{-1} 2$.
35. Evaluate $\csc^{-1} 2$.
36. Find the principal value of $\sin^{-1} \left(\sin \frac{3\pi}{4}\right)$.
37. Find the principal value of $\cos^{-1} \left(\cos \frac{5\pi}{3}\right)$.
38. Find the principal value of $\tan^{-1} \left(\tan \frac{5\pi}{4}\right)$.
39. Evaluate $\sin^{-1} \left(\cos \frac{\pi}{3}\right)$.
40. Evaluate $\cos^{-1} \left(\sin \frac{\pi}{3}\right)$.
41. Evaluate $\sin \left(\sin^{-1} \frac{3}{5}\right)$.
42. Evaluate $\cos \left(\cos^{-1} \frac{4}{5}\right)$.
43. Evaluate $\tan \left(\tan^{-1} \frac{7}{3}\right)$.

44. Evaluate $\sin^{-1}(\sin 5)$ in principal value.
45. Evaluate $\cos^{-1}(\cos 4)$ in principal value.
46. Evaluate $\tan^{-1}(\tan(-2))$ in principal value.
47. Evaluate $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) + \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$.
48. Evaluate $\sin^{-1}\left(\frac{\sqrt{2}}{2}\right) + \cos^{-1}\left(\frac{\sqrt{2}}{2}\right)$.
49. Evaluate $\tan^{-1}(\sqrt{3}) - \tan^{-1}(1)$.
50. Evaluate $\sin^{-1}\left(\frac{3}{5}\right) + \cos^{-1}\left(\frac{4}{5}\right)$.

SECTION 3: BASIC IDENTITIES & PROOFS (Q51-Q80)

51. Prove that $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ for all $x \in [-1, 1]$.
52. Prove that $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$ for all $x > 0$.
53. Show that $\sec^{-1} x + \csc^{-1} x = \frac{\pi}{2}$ for $x > 1$.
54. Prove that $\sin^{-1}(-x) = -\sin^{-1} x$ for all $x \in [-1, 1]$.
55. Prove that $\tan^{-1}(-x) = -\tan^{-1} x$ for all real x .
56. Show that $\cos^{-1}(-x) = \pi - \cos^{-1} x$ for all $x \in [-1, 1]$.
57. Show that $\cot^{-1}(-x) = \pi - \cot^{-1} x$ for all real x .
58. Prove that $\sin^{-1} x = \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$ for $|x| < 1$.
59. Prove that $\cos^{-1} x = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$ for $0 < x \leq 1$.
60. Show that $2\sin^{-1} x = \cos^{-1}(1 - 2x^2)$ for suitable values of x .
61. If $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2}$, show that $x^2 + y^2 = 1$.
62. Prove that $\sin^{-1} x + \sin^{-1} y = \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2})$ under suitable conditions on x and y .
63. Prove that $\sin^{-1} x - \sin^{-1} y = \sin^{-1}(x\sqrt{1-y^2} - y\sqrt{1-x^2})$ under suitable conditions.
64. Prove that $\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$ when $xy < 1$.

65. Prove that $\tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x-y}{1+xy} \right)$ when $xy > -1$.
66. Show that $\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} = \frac{\pi}{4}$.
67. Show that $\tan^{-1} 2 + \tan^{-1} 3 = \frac{\pi}{2}$.
68. Prove that $\sin^{-1} \frac{3}{5} + \cos^{-1} \frac{4}{5} = \frac{\pi}{2}$.
69. Prove that $\sin^{-1} x + \sin^{-1} \sqrt{1-x^2} = \frac{\pi}{2}$ for $x \in [0,1]$.
70. Prove that $\cos^{-1} x + \cos^{-1} \sqrt{1-x^2} = \frac{\pi}{2}$ for $x \in [0,1]$.
71. Show that $\tan^{-1} x + \tan^{-1} \frac{1}{x} = \frac{\pi}{2}$ for $x > 0$.
72. Show that $\cot^{-1} x + \cot^{-1} \frac{1}{x} = \frac{\pi}{2}$ for $x > 0$.
73. Prove that $\sin^{-1} x + \cos^{-1} x$ is independent of x .
74. Prove that $\sin^{-1} x$ is an odd function.
75. Prove that $\cos^{-1} x$ is neither even nor odd.
76. Prove that $\tan^{-1} x$ is an odd function.
77. Show that $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \pi$ if $x^2 + y^2 + z^2 + 2xyz = 1$.
78. Prove that if $\sin^{-1} x = \cos^{-1} y$ and $x, y \in [0,1]$, then $x^2 + y^2 = 1$.
79. Show that if $\sin^{-1} x = \sin^{-1} y$ in principal range, then $x = y$.
80. Show that if $\cos^{-1} x = \cos^{-1} y$ in principal range, then $x = y$.

SECTION 4: COMPOSITION WITH TRIG FUNCTIONS (Q81-Q110)

81. Find $\sin (\sin^{-1} x)$.
82. Find $\cos (\cos^{-1} x)$.
83. Find $\tan (\tan^{-1} x)$.
84. Express $\sin (\cos^{-1} x)$ in terms of x .
85. Express $\cos (\sin^{-1} x)$ in terms of x .
86. Express $\tan (\sin^{-1} x)$ in terms of x .
87. Express $\sin (\tan^{-1} x)$ in terms of x .

88. Express $\cos (\tan^{-1} x)$ in terms of x .
89. Express $\cot (\tan^{-1} x)$ in terms of x .
90. If $\theta = \sin^{-1} \frac{3}{5}$, find $\cos \theta$.
91. If $\theta = \cos^{-1} \frac{4}{5}$, find $\sin \theta$.
92. If $\theta = \sin^{-1} \frac{12}{13}$, find $\tan \theta$.
93. If $\theta = \tan^{-1} \frac{3}{4}$, find $\sin \theta$.
94. If $\theta = \tan^{-1} \frac{3}{4}$, find $\cos \theta$.
95. If $\theta = \sin^{-1} \frac{5}{13}$, find $\sec \theta$.
96. If $\theta = \cos^{-1} \frac{5}{13}$, find $\csc \theta$.
97. Express $\sin^{-1} \left(\frac{2t}{1+t^2} \right)$ in terms of $\tan^{-1} t$.
98. Express $\cos^{-1} \left(\frac{1-t^2}{1+t^2} \right)$ in terms of $\tan^{-1} t$.
99. Express $\tan^{-1} \left(\frac{2x}{1-x^2} \right)$ in terms of $\tan^{-1} x$.
100. Simplify $\sin^{-1} (\sqrt{1 - \cos^2 x})$.
101. Simplify $\cos^{-1} (\sqrt{1 - \sin^2 x})$.
102. If $0 < x < \frac{\pi}{2}$, express $\sin^{-1} (\sin 2x)$ in terms of x .
103. If $0 < x < \frac{\pi}{2}$, express $\cos^{-1} (\cos 2x)$ in terms of x .
104. If $0 < x < \frac{\pi}{2}$, find $\sin (\cos^{-1} (\sin x))$.
105. If $0 < x < \frac{\pi}{2}$, find $\cos (\sin^{-1} (\cos x))$.
106. Express $\sin^{-1} (2 \sin x \cos x)$ in terms of x using double-angle formula.
107. Express $\cos^{-1} (\cos^2 x - \sin^2 x)$ in terms of x .
108. If $\theta = \tan^{-1} x$, show that $\sin (2\theta) = \frac{2x}{1+x^2}$.
109. If $\theta = \tan^{-1} x$, show that $\cos (2\theta) = \frac{1-x^2}{1+x^2}$.
110. If $\theta = \sin^{-1} x$, show that $\cos (2\theta) = 1 - 2x^2$.

SECTION 5: SIMPLE EQUATIONS & INEQUALITIES (Q111–Q150)

111. Solve $\sin^{-1} x = \frac{\pi}{6}$.

112. Solve $\cos^{-1} x = \frac{2\pi}{3}$.

113. Solve $\tan^{-1} x = \frac{\pi}{4}$.

114. Solve $\sin^{-1} x = \cos^{-1} \frac{1}{2}$.

115. Solve $\sin^{-1} x + \sin^{-1} x = \frac{\pi}{2}$.

116. Solve $\cos^{-1} x + \cos^{-1} x = \pi$.

117. Find all $x \in [-1, 1]$ such that $\sin^{-1} x \leq \frac{\pi}{4}$.

118. Find all $x \in [-1, 1]$ such that $\cos^{-1} x \geq \frac{\pi}{3}$.

119. Solve $\tan^{-1} x + \tan^{-1} x = \frac{\pi}{4}$.

120. Solve $\sin^{-1} (2x) = \frac{\pi}{6}$ for x .

121. Solve $\cos^{-1} (2x^2 - 1) = \frac{\pi}{3}$ for x .

122. Solve $\tan^{-1} \left(\frac{2x}{1-x^2} \right) = \frac{\pi}{4}$.

123. Solve $\sin^{-1} x = \sin^{-1} \frac{3}{5}$.

124. Solve $\cos^{-1} x = \cos^{-1} \frac{4}{5}$.

125. Solve $\tan^{-1} x = \tan^{-1} \frac{7}{3}$.

126. Find all x such that $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$.

127. Solve $\sin^{-1} (1 - 2x^2) = 0$.

128. Solve $\cos^{-1} (1 - 2x^2) = \frac{\pi}{2}$.

129. Solve $\tan^{-1} x + \tan^{-1} \frac{1-x}{1+x} = \frac{\pi}{4}$.

130. Find the range of the function $f(x) = \sin^{-1} (2x - 1)$.

131. Find the range of the function $f(x) = \cos^{-1} (2x - 1)$.

132. Find the range of the function $f(x) = \tan^{-1} (2x)$.

133. Determine the domain of $f(x) = \sin^{-1} \left(\frac{1-x}{1+x} \right)$.
134. Determine the domain of $f(x) = \cos^{-1} \left(\frac{2x}{1+x^2} \right)$.
135. Determine the domain of $f(x) = \tan^{-1} \left(\frac{1-x^2}{1+x^2} \right)$.
136. Solve the inequality $\sin^{-1} x > 0$.
137. Solve the inequality $\cos^{-1} x < \frac{\pi}{2}$.
138. Solve the inequality $\tan^{-1} x \geq 0$.
139. Solve the inequality $\sin^{-1} (2x - 1) \in \left[0, \frac{\pi}{2} \right]$.
140. Solve the inequality $\cos^{-1} (2x - 1) \in [0, \pi]$.
141. Find all x such that $\sin^{-1} x = \cos^{-1} x$.
142. Find all x such that $\sin^{-1} x + \sin^{-1} (1 - x) = \frac{\pi}{2}$.
143. Find all x such that $\sin^{-1} x + \cos^{-1} (1 - x) = \frac{\pi}{2}$.
144. Solve $\sin^{-1} x = \tan^{-1} x$ in principal ranges.
145. Solve $\cos^{-1} x = \tan^{-1} x$ in principal ranges.
146. For which values of x is $\sin^{-1} x$ an increasing function?
147. For which values of x is $\cos^{-1} x$ a decreasing function?
148. Show that $\sin^{-1} x$ and $\cos^{-1} x$ are continuous on $[-1, 1]$.
149. Show that $\tan^{-1} x$ is continuous on $(-\infty, \infty)$.
150. Comment on the monotonicity of $\tan^{-1} x$ on $\mathbb{R}$.